Moffat Bay Lodge ERD Description

Moffat Bay Lodge Database Design Specification

Section 1: Executive Summary

Project Overview

Moffat Bay Lodge is a web-based reservation management system designed to provide visitors with information about the lodge, room options, pricing, amenities, local attractions, and reservation services. The system allows guests to browse lodge information without registration while requiring account creation and authentication before reservations can be made.

The database serves as the foundation of the application and supports all major functional requirements identified in the Technical Design Document (TDD). The design follows relational database principles and Third Normal Form (3NF) normalization standards to ensure data integrity, minimize redundancy, and support efficient retrieval of customer, room, and reservation data.

Project Objectives

This database supports:

- Customer registration
- Customer authentication
- Session management
- Room information management
- Room type management
- Amenity management
- Reservation creation
- Reservation confirmation
- Reservation lookup
- Attraction information display
- Lodge information management
- Image management

Section 2: Core Business Objects

Business Object 1: Customer

The Customer represents any person who registers with Moffat Bay Lodge. Customers create accounts, authenticate themselves, create reservations, and review reservation information.

Responsibilities

- Account registration
- User authentication
- Reservation ownership
- Reservation lookup
- Communication with lodge staff

Business Object 2: Reservation

The Reservation represents the core transaction of the business.

A reservation connects

Customer
Room
Reservation Dates
Guests
Status

Reservations record:

- Check-in date
- Check-out date
- Number of guests
- Special requests
- Cost calculations

Business Object 3: Room

Rooms represent the physical accommodation available for booking.

Examples:

Plain Text
Room 101
Room 102
Room 201
Room 202

Every room is associated with:

- A room type
- A room status
- Amenities
- Reservations

Business Object 4: Room Type

Room types define pricing and capacity.

Examples:

King
Queen
Double Queen
Double Full

The room type determines:

- Price per night
- Maximum guests
- Number of beds

Business Object 5: Amenity

Amenities represent features available within rooms.

Examples:

- Wi-Fi
- Television
- Fireplace

- Balcony
- Refrigerator

Business Object 6: Attraction

Attractions provide informational content for visitors.

Examples:

- Hiking
- Whale Watching
- Fishing
- Kayaking

Business Object 7: Lodge

The lodge acts as the central organizational entity.

The lodge manages:

- Rooms
- Amenities
- Attractions
- Images
- Contact information

Section 3: Complete ERD Coverage

The ERD fully supports all user stories from the TDD.

User Story 01: Customer Registration

Supported Entities:

CUSTOMER

CUSTOMER_ADDRESS

Supported Functions:

- New account creation
- Email uniqueness verification
- Password storage
- Contact information collection

User Story 02: Customer Login

Supported Entities:

CUSTOMER
LOGIN_SESSION

Supported Functions:

- User authentication
- Login validation
- Session management
- Logout functionality

User Story 03: Room Reservation

Supported Entities:

Plain Text
ROOM
ROOM_TYPE
RESERVATION

Supported Functions:

- Room selection
- Guest count validation
- Date validation
- Reservation creation

User Story 04: Reservation Summary

Supported Entities:

RESERVATION
ROOM
ROOM_TYPE

Supported Functions:

- Reservation review
- Cost calculation
- Reservation confirmation

User Story 05: Reservation Lookup

Supported Entities:

RESERVATION
RESERVATION_LOOKUP
CUSTOMER

Supported Functions:

- Lookup by email
- Lookup by phone
- Lookup by reservation number

User Story 06: Room Information

Supported Entities:

ROOM
ROOM_TYPE
AMENITY

Supported Functions:

- View room details
- View room pricing
- View amenities

User Story 07: Attractions

Supported Entities:

ATTRACTION

Supported Functions:

- Browse nearby attractions
- Vacation Planning

User Story 08: Reservation Date Review

Supported Entities:

RESERVATION

Supported Functions:

- View check-in date
- View check-out date

User Story 09: Lodge Information

Supported Entities:

LODGE_INFO

Supported Functions:

- Contact information

- Business hours
- Address display

Section 4: Entity Documentation

Entity 1: CUSTOMER

Purpose

Stores information about registered customers.

Attributes

Attribute	Data Type
customer_id	INT
first_name	VARCHAR(50)
last_name	VARCHAR(50)
email	VARCHAR(100)
phone_number	VARCHAR(20)
password_hash	VARCHAR(255)
registration_date	DATETIME
last_login_date	DATETIME
account_status	VARCHAR(20)

Business Rules

1. Must have unique email address.
2. Passwords must be encrypted.
3. Customer must register before booking.
4. Customer may own multiple reservations.
5. Customer may have multiple login sessions.
6. Customer must have active account.
7. Customer cannot share email address.
8. Registration date cannot be null.
9. Customer data may be updated.

10. Customer may have multiple addresses.

Example Records

1. John Smith
2. Maria Garcia
3. Emily Davis
4. Sarah Brown
5. David Wilson
6. James Anderson
7. Michael Johnson
8. Sophia Martinez
9. Daniel Thomas
10. Kevin Clark

Entity 2: CUSTOMER_ADDRESS

Purpose

Stores address information associated with customers.

Attributes

Attribute	Data Type
address_id	INT
customer_id	INT
street_address	VARCHAR(150)
city	VARCHAR(50)
state	VARCHAR(50)
zip_code	VARCHAR(15)
country	VARCHAR(50)

Business Rules

1. Address must belong to a customer.
2. Customer may have multiple addresses.

3. Address cannot exist without customer.
4. Country is required for international customers.
5. Postal codes vary by country.

Example Records

1. 123 Main Street
2. 456 Oak Drive
3. 789 Pine Avenue
4. 100 River Road
5. 44 Harbor Lane

Entity 3: LOGIN_SESSION

Purpose

Tracks user authentication sessions.

Attributes

Attribute	Data Type
session_id	INT
customer_id	INT
login_time	DATETIME
logout_time	DATETIME
session_token	VARCHAR(255)
ip_address	VARCHAR(45)
session_status	VARCHAR(20)

Example Session Statuses

1. Active
2. Expired
3. LoggedOut
4. Timeout
5. Invalidated

6. Locked
7. Failed
8. Reauthenticated
9. Suspended
10. Terminated

Entity 4: ROOM_TYPE

Purpose

Defines categories of rooms and pricing structures.

Attributes

Attribute	Data Type
room_type_id	INT
room_type_name	VARCHAR(50)
description	TEXT
price_per_night	DECIMAL(10,2)
max_guests	INT
bed_count	INT

Example Records

1. King
2. Queen
3. Double Queen
4. Double Full
5. Family Suite
6. Deluxe King
7. Honeymoon Suite
8. Luxury Suite
9. Ocean View King
10. Mountain View Queen

Entity 5: ROOM

Purpose

Represents physical rooms available at the lodge.

Attributes

Attribute	Data Type
room_id	INT
lodge_id	INT
room_type_id	INT
room_status_id	INT
room_number	VARCHAR(10)
floor_number	INT
view_type	VARCHAR(50)
notes	TEXT

Example Rooms

101, 102, 103, 104, 105, 201, 202, 203, 204, 205

Entity 6: ROOM_STATUS

Purpose

Tracks room availability.

Example Values

1. Available
2. Occupied
3. Reserved
4. Maintenance
5. Cleaning

6. Out Of Service
7. Ready
8. Pending Inspection
9. Blocked
10. Renovation

Entity 7: RESERVATION

Purpose

Stores customer bookings.

Attributes

Attribute	Data Type
reservation_id	INT
customer_id	INT
room_id	INT
status_id	INT
reservation_date	DATETIME
check_in_date	DATE
check_out_date	DATE
guest_count	INT
total_nights	INT
total_cost	DECIMAL(10,2)
special_requests	TEXT

Business Rules

1. Customer required.
2. Room required.
3. Status required.
4. Check-out must be later than check-in.
5. Guest count must not exceed room capacity.
6. Total cost calculated automatically.

7. Reservations cannot overlap.
8. Reservation must be reviewed before confirmation.
9. Must have reservation date.
10. Must belong to one customer.

Example Reservations

R1001-R1010

Entities 8-15 Summary

RESERVATION_STATUS

Examples:

Pending, Confirmed, Cancelled, Completed, Checked In, Checked Out, No-show, Refunded, Modified, Expired

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RESERVATION_GUEST

Stores all guests attached to a reservation.

Examples:

John Smith, Mary Smith, Sarah Smith, Tim Smith, Emma Lee, etc.

RESERVATION_LOOKUP

Tracks searches using:

- Email
- Phone
- Reservation ID

Examples:

Lookup001-Lookup010

AMENITY

Examples:

- WiFi
- TV
- Fireplace
- Balcony
- Refrigerator
- Coffee Maker
- Ocean View
- Mountain View
- Jacuzzi
- Microwave

ROOM_AMENITY

Examples:

- Room101-WiFi
- Room101-TV
- Room101-Fireplace
- Room102-WiFi
- Room102-TV
- Room103-Balcony
- Room104-Jacuzzi
- Room105-Refrigerator
- Room201-OceanView
- Room202-MountainView

ATTRACTION

Examples:

- Whale Watching
- Fishing
- Hiking
- Kayaking
- Beach Access
- Boat Tours
- Waterfall Trail
- Nature Preserve
- Local Museum
- Fine Dining

LODGE_INFO

Examples:

- Main Lodge
- Reservations Office
- Front Desk
- Management Office
- Customer Support

LODGE_IMAGE

Examples:

- Main Building
- Lobby
- Deluxe Suite
- Restaurant
- Hiking Trail
- Beach View
- Sunset Photo
- Fishing Dock
- Balcony View
- Whale Watching Tour

Section 5: Assumptions and Constraints

Database Constraints

1. Email addresses are unique.
2. Passwords are hashed.
3. Registration required before booking.
4. No online payment processing.
5. Room numbers must be unique.
6. Room type determines price.
7. Check-in date required.
8. Check-out date required.
9. Guest count required.
10. Room capacity enforced.
11. Active account required.
12. Session token unique.
13. Reservation date required.
14. Status required.
15. Reservation linked to room.
16. Reservation linked to customer.
17. Room must belong to lodge.
18. Attraction belongs to lodge.
19. Images belong to lodge.
20. Amenities can be reused.
21. Foreign key integrity enforced.
22. No orphan records allowed.
23. Duplicate room-amenity pairs are prohibited.
24. Reservation overlaps prohibited.
25. System follows 3NF normalization.
26. Lodge information visible publicly.
27. Attractions visible publicly.
28. Reservation data private.
29. Customer authentication required.
30. Customer ownership controls reservation access.

Section 6: Normalization Analysis

First Normal Form (1NF)

- No repeating groups
- Atomic attributes
- Unique records

Second Normal Form (2NF)

- Non-key attributes depend on entire primary key
- No partial dependencies

Third Normal Form (3NF)

- No transitive dependencies
- Customer data stored separately from reservations
- Room pricing stored in Room Type
- Amenity data stored separately from rooms

Conclusion

The proposed entities Moffat Bay Lodge database design fully supports all user stories, business requirements, and future system expansion needs. The design implements strong normalization principles, proper primary and foreign key relationships, reservation management workflows, authentication support, room and amenity management, attraction tracking, and lodge content management, making it suitable for implementation using Java, JSP, MySQL, HTML, PHP and CSS.